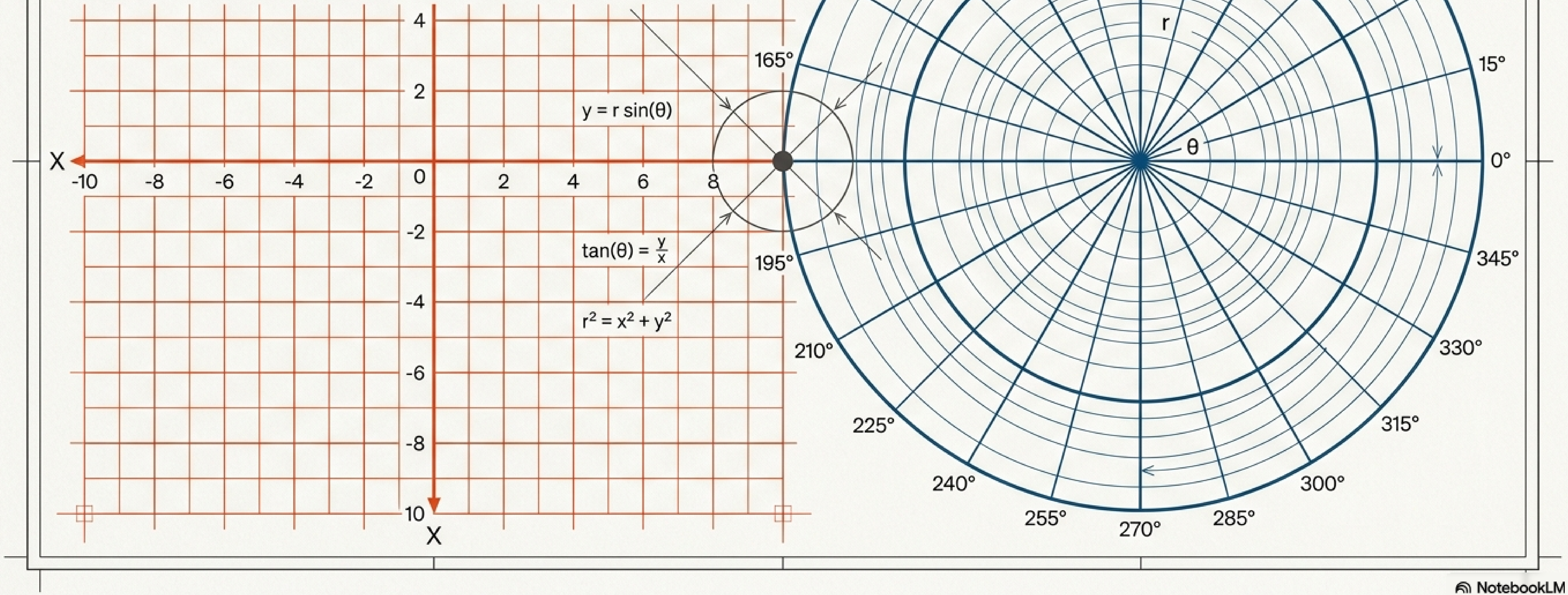
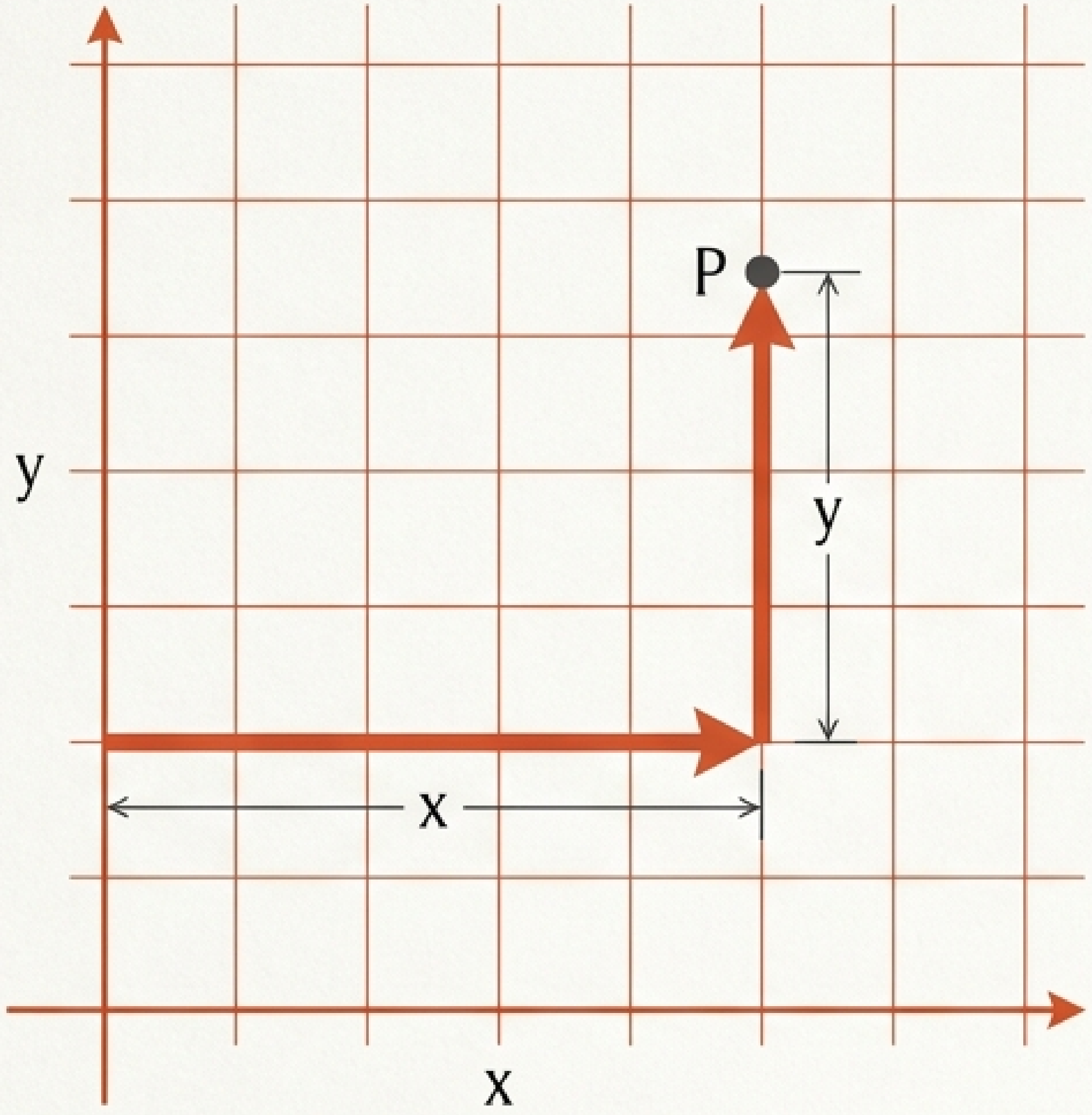
Bridging Two Worlds

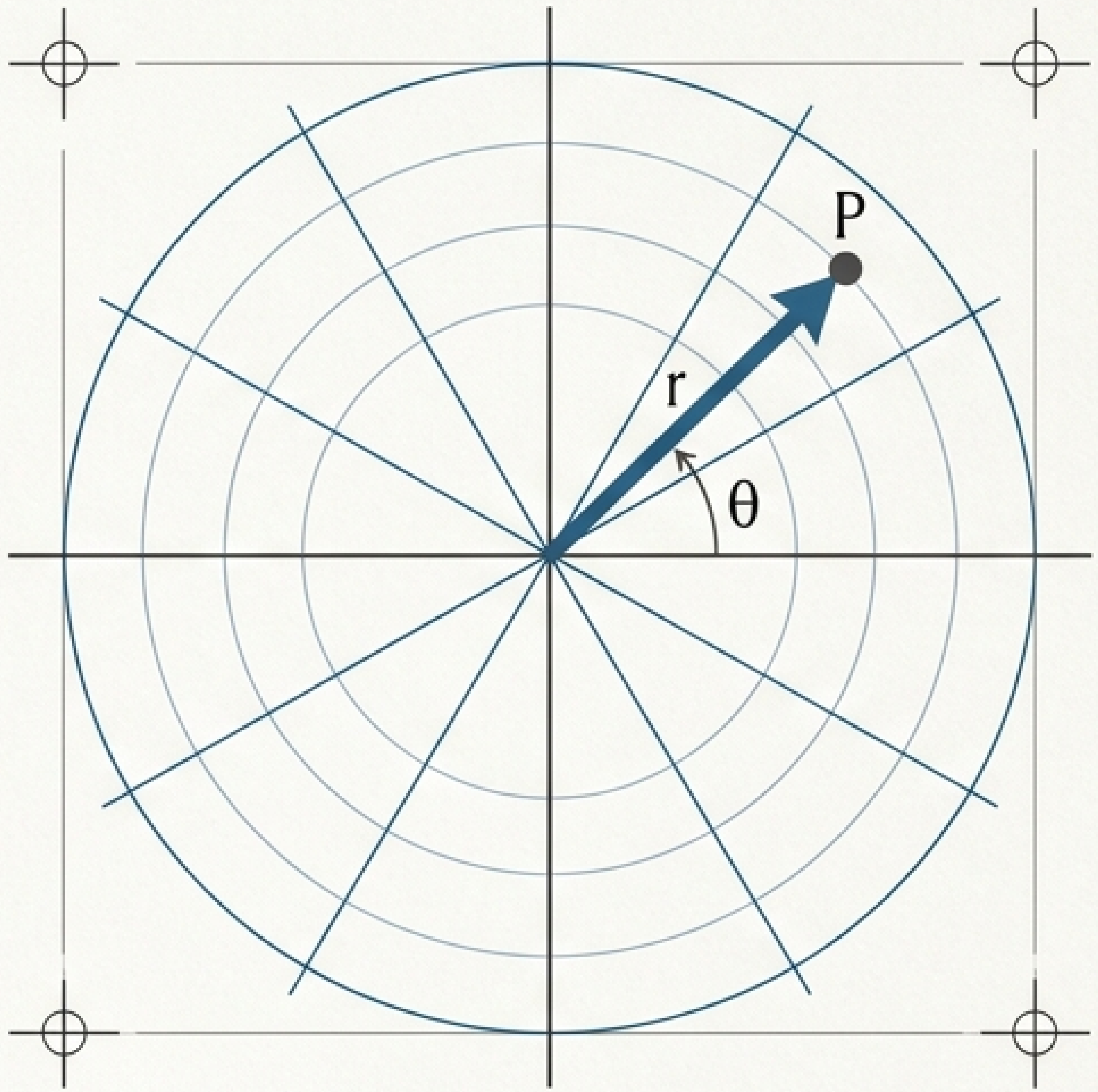
A Visual Guide to Translating Between Polar and Rectangular Coordinate Systems





The City Grid

Navigating via horizontal and vertical distances (x, y).



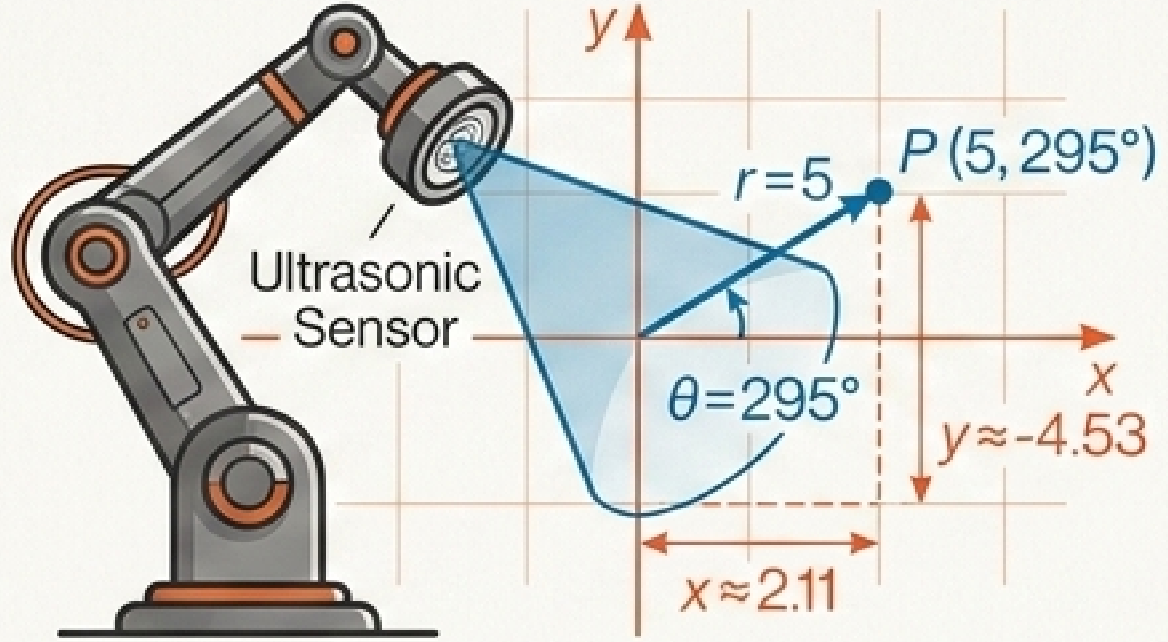
The Radar Sweep

Navigating via distance and direction (r, θ).

Both systems identify the exact same point P in space. The challenge is seamlessly translating between them.

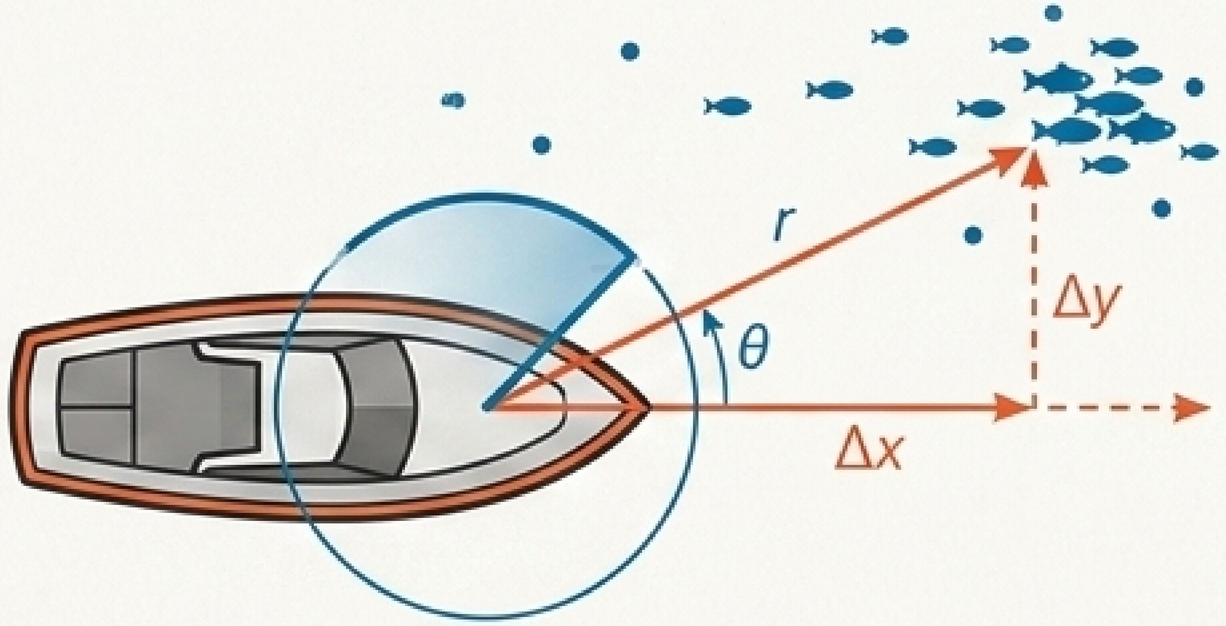
Why Translate?

The Robotics Problem



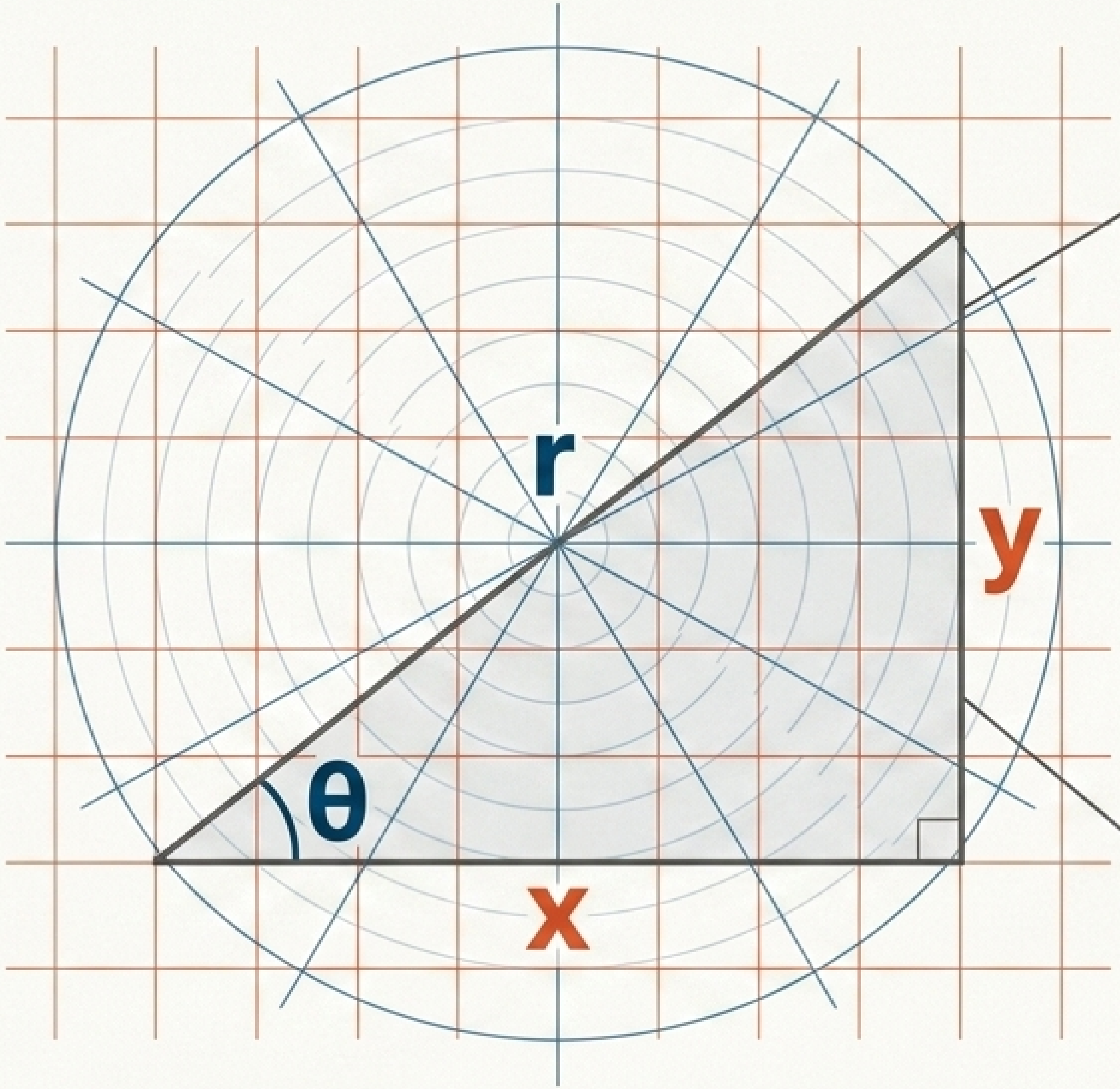
A robot's native language is Polar (sweeping sensors), but its internal spatial map requires Rectangular coordinates.

The Marine Radar Problem



Radar tracks via radius and angle, but plotting a linear intercept course demands Cartesian translation.

The Translation Key: The Foundational Triangle



$\cos\theta = \frac{x}{r}$

\sin $\theta = \frac{y}{r}$

Every translation between these two worlds is governed by the geometry of this single right triangle.

Translating Points [Polar to Rectangular]

The Forward Path

The Input

$$(4, \pi/6)$$

The Translation Engine

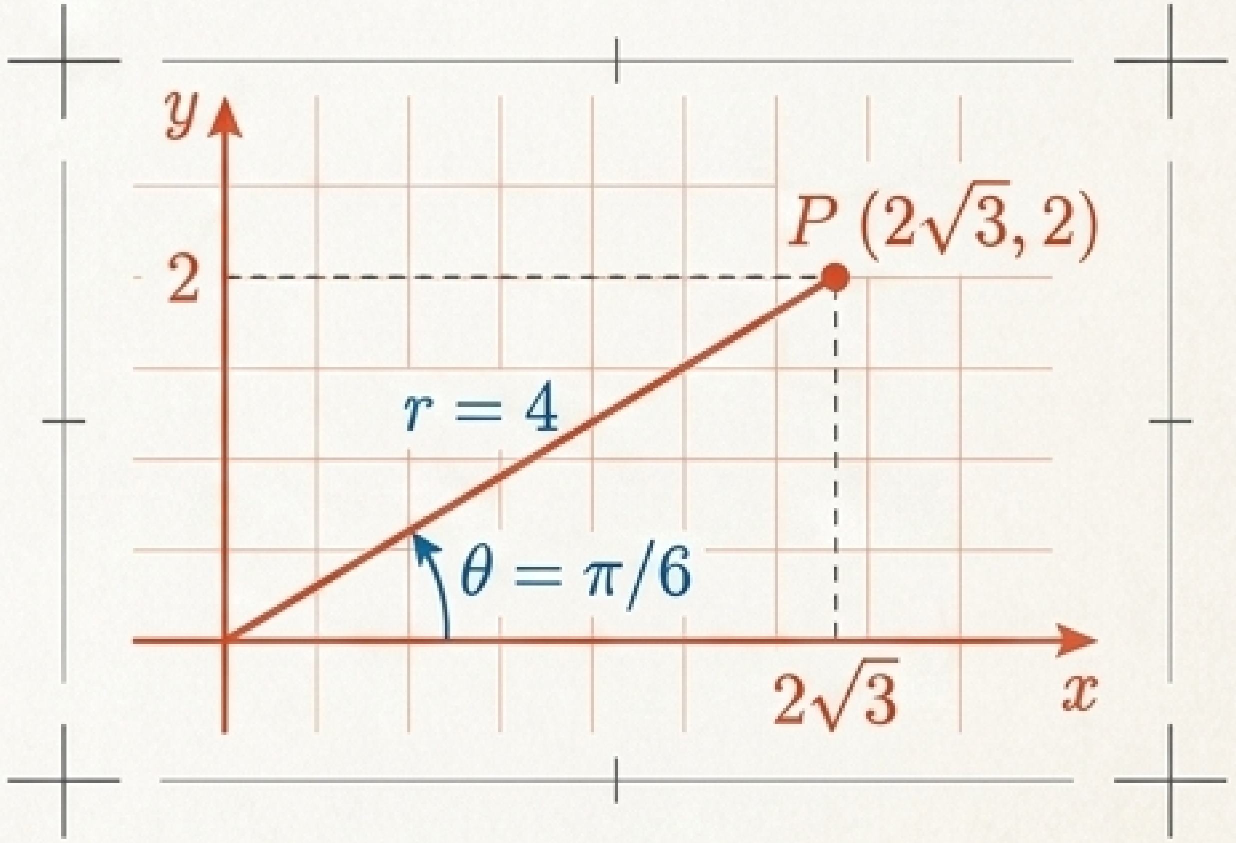
$$x = r\cos\theta$$

$$y = r\sin\theta$$

The Output

$$x = 4\cos(\pi/6) \to 4(\sqrt{3}/2) \to 2\sqrt{3}$$

$$y = 4\sin(\pi/6) \to 4(1/2) \to 2$$

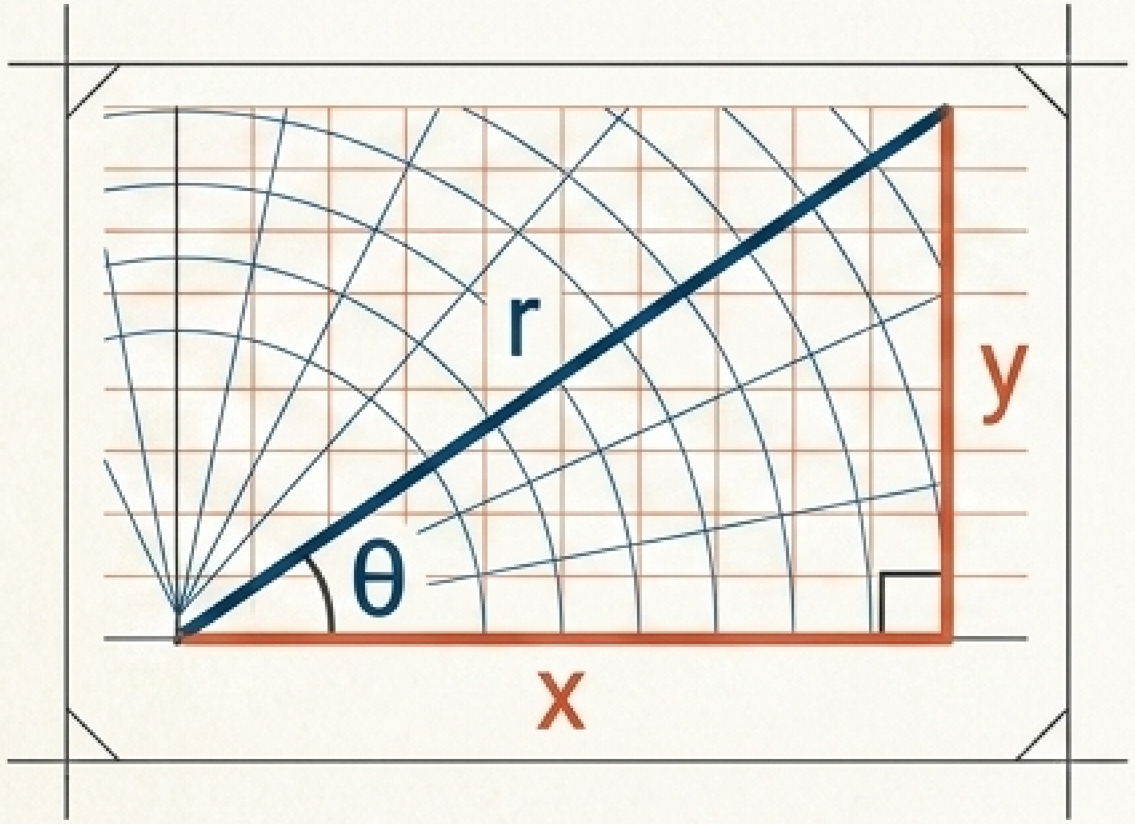


Translating Points [Rectangular to Polar]

Finding Distance

$$r = \sqrt{x^2 + y^2}$$

The pure geometric certainty of the Pythagorean theorem.

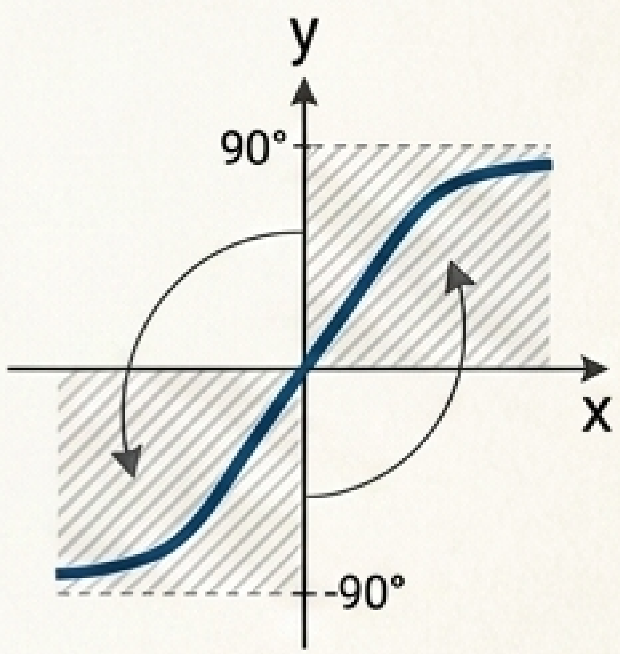


The Direction Challenge

$$\theta = \tan^{-1}(y/x)$$

The ArcTangent Blind Spot:

Inverse tangent only outputs values between -90° and 90°. If your point lies on the left side of the map (where x is negative), the mathematical formula requires manual navigation.



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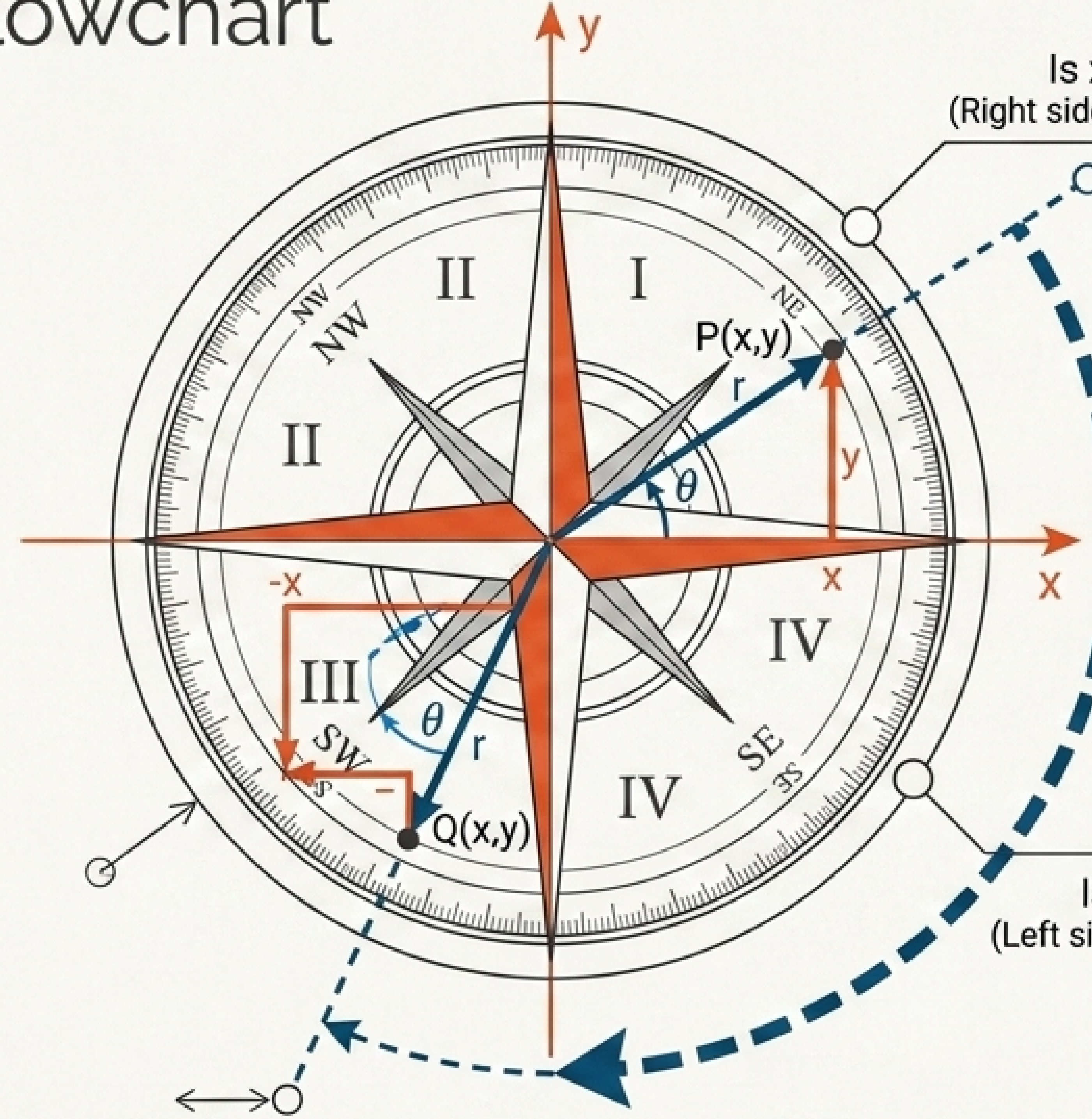
Navigating the Quadrants: The θ Correction Flowchart



Is x > 0?
(Right side of the map)

Action:
Trust the formula.

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

Is x < 0?
(Left side of the map)

Action:
The formula points backward. Add a half-circle correction.

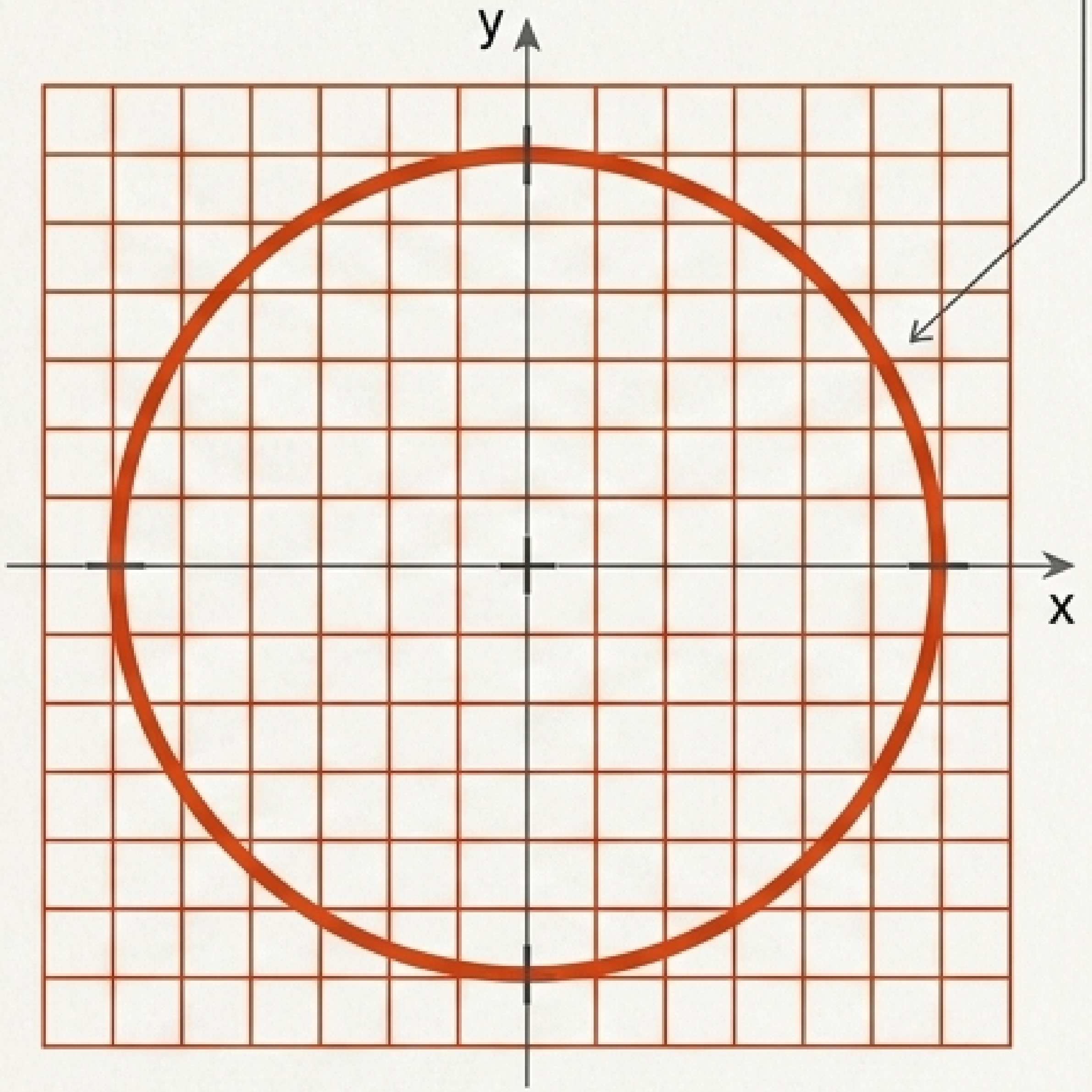
$$\theta = \tan^{-1}\left(\frac{y}{x}\right) + \pi$$

Point Conversion Diagnostic Matrix

Original Point	Pathway & Rule	Translated Coordinates
$(-2, 135^\circ)$	Cartesian Path: $x = r\cos\theta, y = r\sin\theta$	$(\sqrt{2}, -\sqrt{2})$
$(3, -120^\circ)$	Cartesian Path: $x = r\cos\theta, y = r\sin\theta$	$(-3/2, -3\sqrt{3}/2)$
$(1, -\sqrt{3})$	Polar Path: Quadrant IV ($x > 0$)	$(2, -\pi/3)$
$(-3, 6)$	Polar Path: Quadrant II ($x < 0$, add π)	$(\approx 6.71, 2.03\text{ rad})$

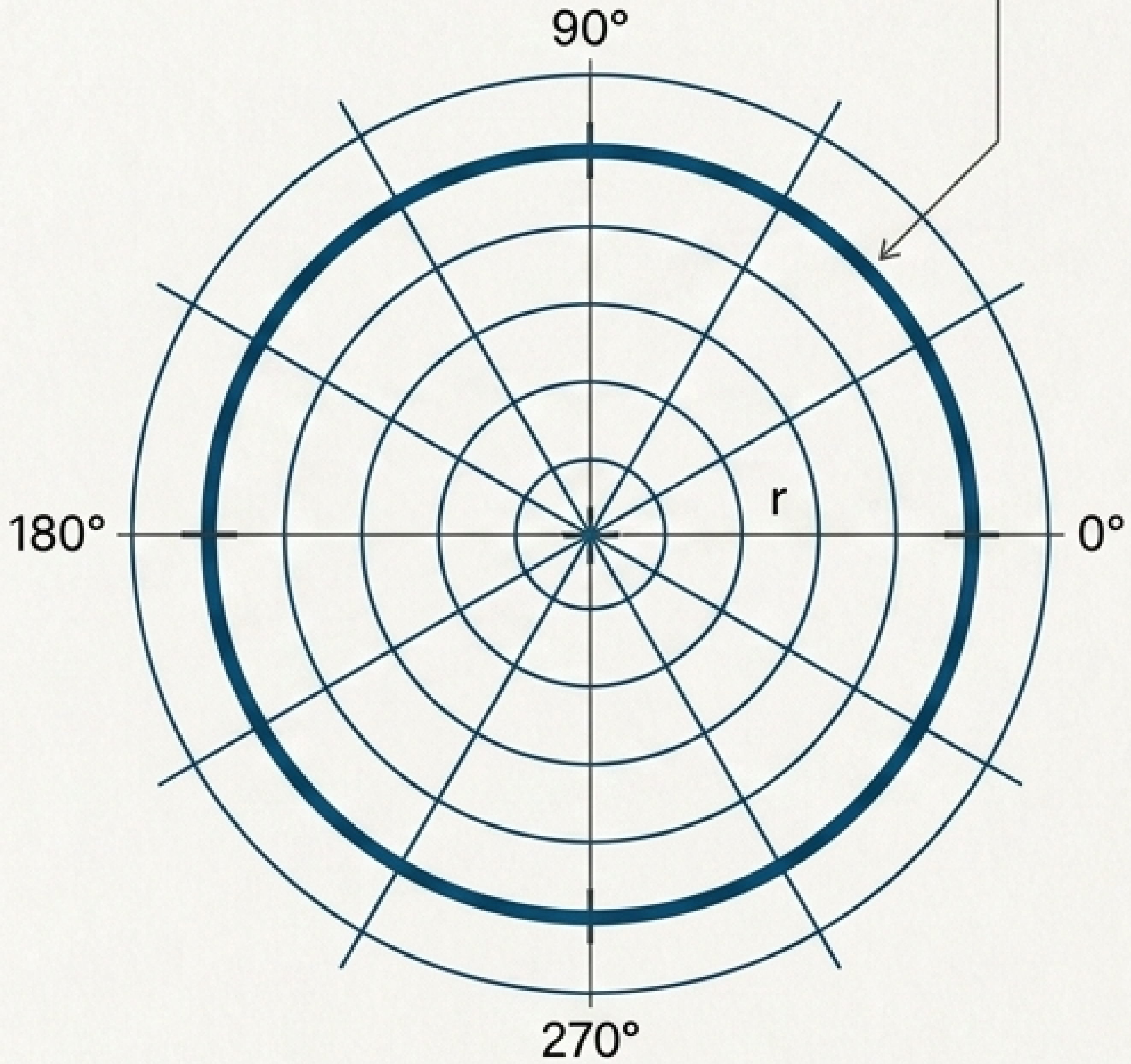
The Cartesian Struggle

$$x^2 + y^2 = 9$$



The Polar Elegance

$$r = 3$$



By translating entire equations, we can turn highly complex algebra into pure, simple geometry.

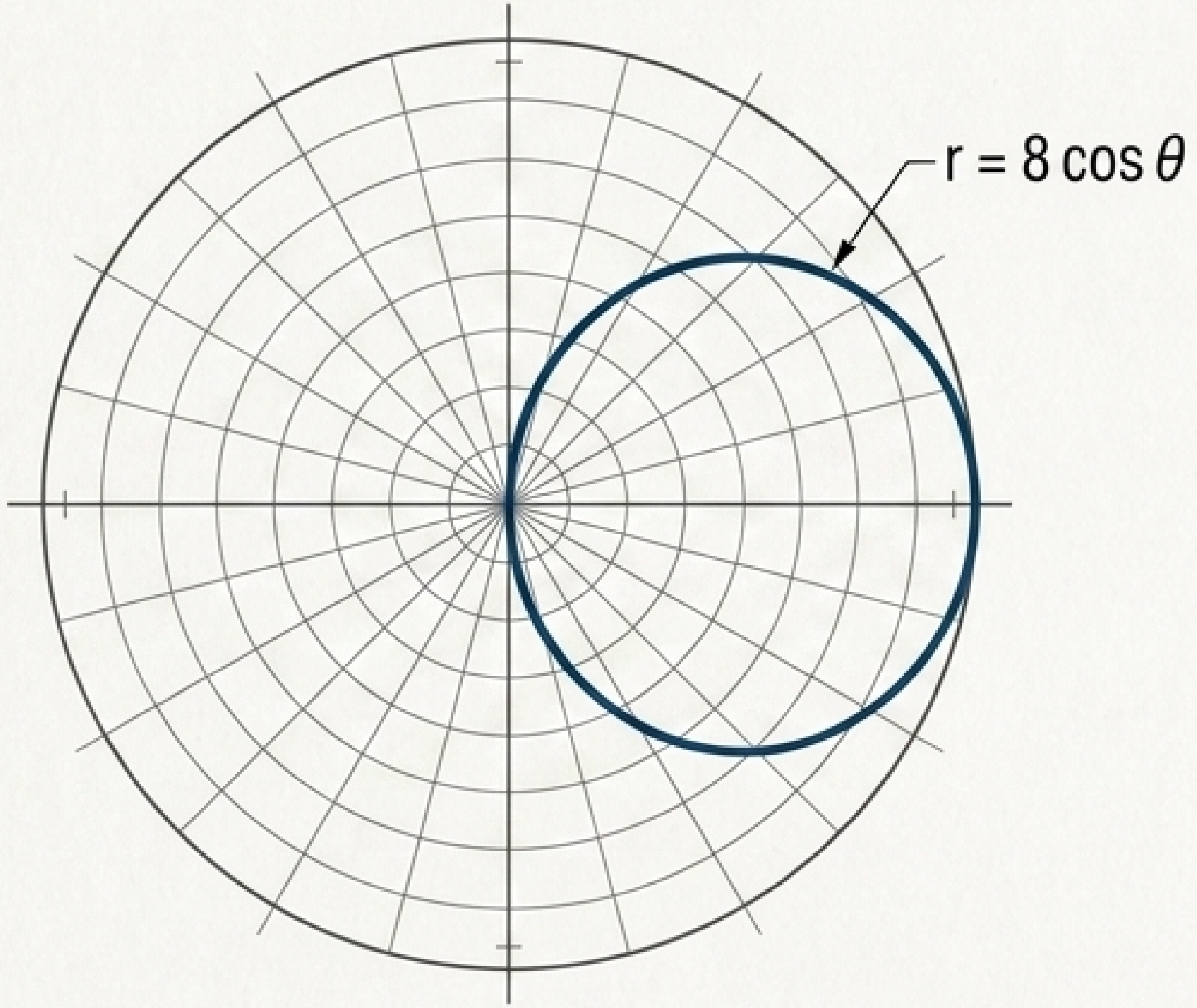
Translating Curves [Rectangular to Polar]

The Offset Circle

$(x - 4)^2 + y^2 = 16$

Expand and substitute x with $r \cos \theta$ and y^2 with $r^2 \sin^2 \theta$

$r = 8 \cos \theta$

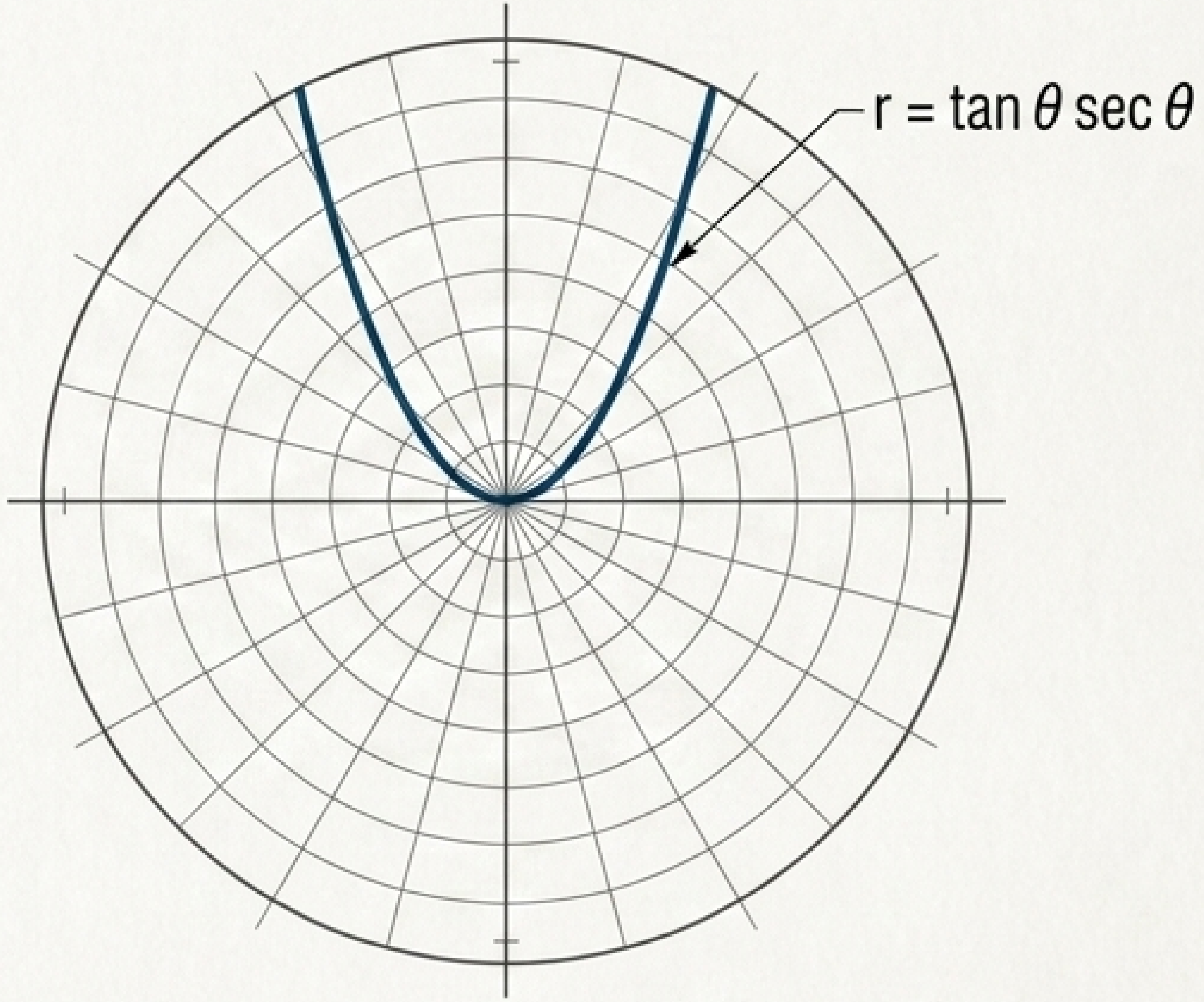


The Parabola

$y = x^2$

Substitute y and x, then isolate r

$r = \tan \theta \sec \theta$



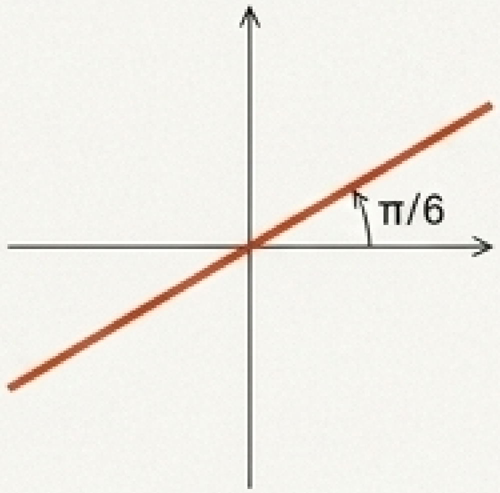
Decoding the Geometry [Polar to Rectangular]

Unlocking a Line

Start
$\theta = \pi/6$

Operation
Take tangent of both sides, substitute $\tan\theta = y/x$

Result
$y = (\frac{\sqrt{3}}{3})x$

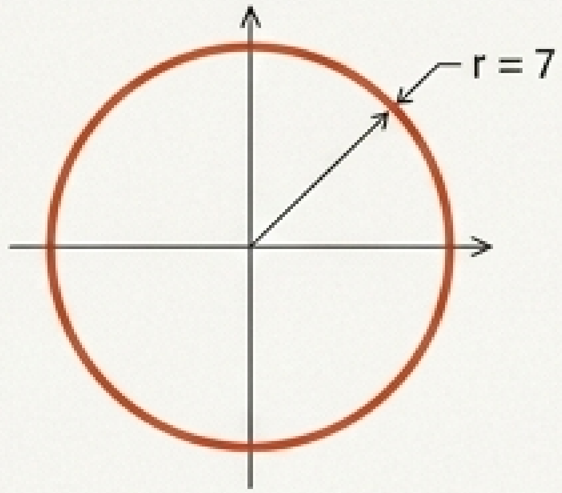


Unlocking a Circle

Start
$r = 7$

Operation
Square both sides, substitute r^2

Result
$x^2 + y^2 = 49$

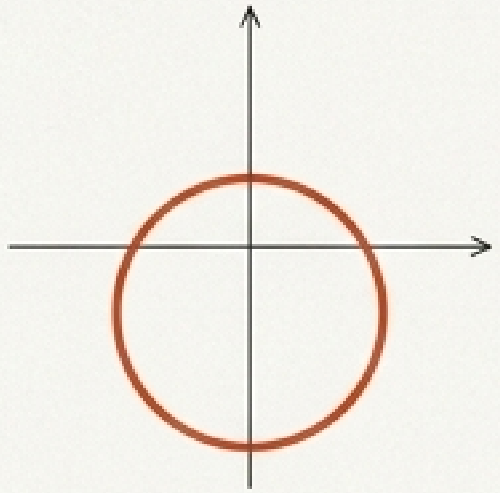


Unlocking an Offset Circle

Start
$r = -5\sin\theta$

Operation
Multiply by r, complete the square

Result
$x^2 + (y + 2.5)^2 = 6.25$



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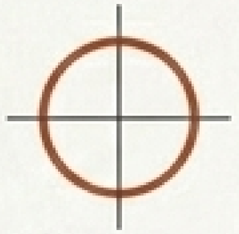
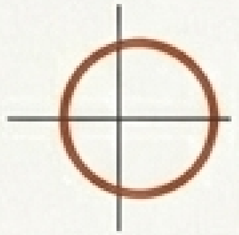
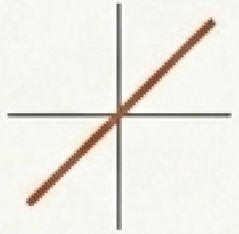
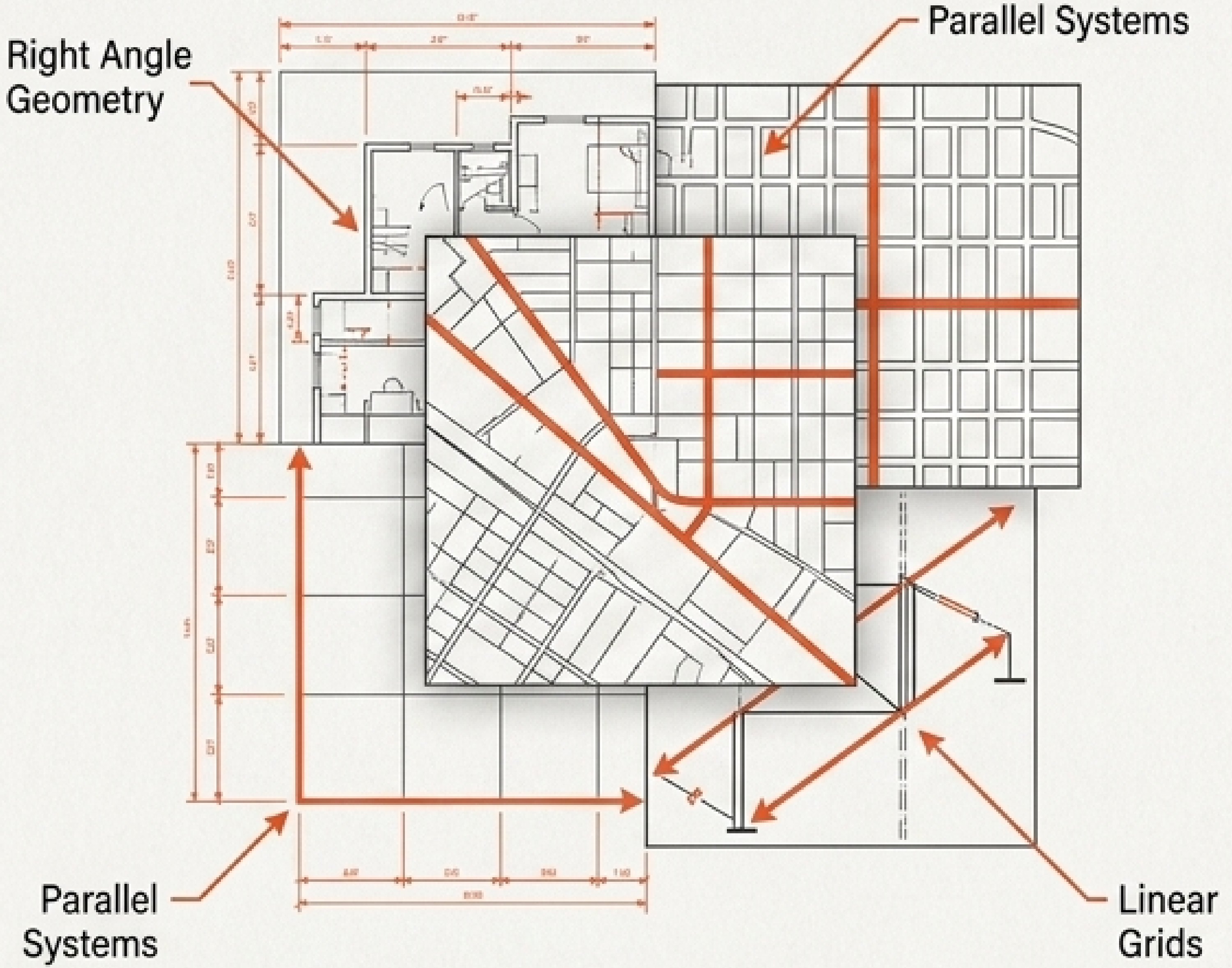
The Shape-Shifter Matrix

Physical Shape	Rectangular Equation	Polar Equation
Circle at Origin 	$x^2 + y^2 = a^2$	$r = a$
Circle on X-Axis 	$(x - a)^2 + y^2 = a^2$	$r = 2a\cos\theta$
Line Through Origin 	$y = mx$	$\theta = c$

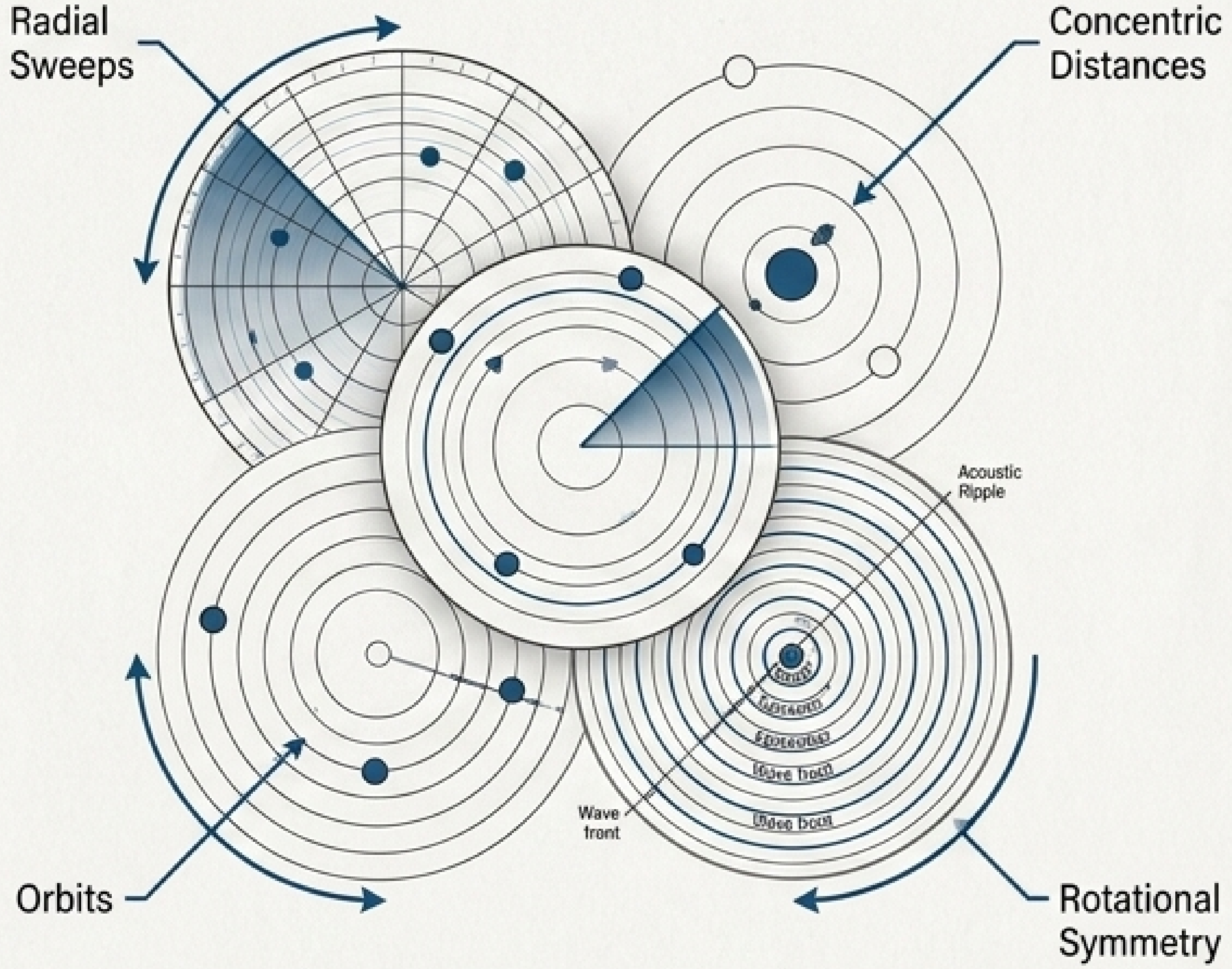
Synthesis: Choosing Your Native Language

Optimize for Linear Space



Use Rectangular when designing straight lines, right angles, and parallel systems.

Optimize for Rotational Space



Use Polar when analyzing sweeps, distances from a central point, orbits, and ripples.

The Navigator's Blueprint: Translation Master Key

Point Conversions [Polar to Rectangular]

$$x = r\cos\theta$$

$$y = r\sin\theta$$

Point Conversions [Rectangular to Polar]

$$r = \sqrt{x^2 + y^2}$$

$$\tan\theta = \frac{y}{x}$$

Rule: Add π if $x < 0$

Equation Conversions

- To build Polar: Substitute x and y.
- To build Rectangular: Create r^2 or substitute trigonometric ratios.

The Foundational Triangle